

Rihua Van Steenburgh

AI Engineer | RAG & Retrieval Systems| Applied NLP

Alpharetta, GA | 470-295-0545 | rihuavan@gmail.com

github.com/rihua-tech | linkedin.com/in/rihua | rihua.dev

Summary

AI Engineer with hands-on project experience building RAG systems, retrieval pipelines, NLP applications, and AI-enabled data products. Skilled in Python, FastAPI, PostgreSQL/pgvector, embeddings, hybrid retrieval, Docker, and CI/CD, with experience in grounded generation, citation validation, RAG evaluation, PyTorch, Transformers, and cloud data platforms.

Skills

- **Languages:** Python, SQL, R, JavaScript, TypeScript
- **AI / RAG & Retrieval:** RAG, embeddings, vector search, pgvector, hybrid retrieval, RRF, retrieval evaluation, grounded generation
- **AI Application & DevOps:** FastAPI, Docker, GitHub Actions, Streamlit, Next.js, Git/GitHub
- **ML / NLP:** scikit-learn, TF-IDF, Linear SVM, PyTorch, Transformers, DistilBERT
- **Data & Processing:** PostgreSQL, PySpark, Delta Lake, dbt, Airflow, dimensional modeling
- **Cloud & Data Platforms:** Azure Data Factory, ADLS Gen2, Databricks, AWS S3, ECS/Fargate, Redshift Serverless

Projects

1. CivicLens RAG Operations Copilot | Python • FastAPI • PostgreSQL/pgvector • Hybrid Retrieval • Docker

May 2026 - Aug 2026 [GitHub](#) | [Live Demo](#) | [Case Study](#)

- Built an end-to-end RAG system with manifest-validated ingestion, section-aware chunking, PostgreSQL/pgvector embeddings, and hybrid semantic + full-text retrieval fused with RRF.
- Deployed a Next.js → Render FastAPI → Neon PostgreSQL/pgvector AI application with grounded generation, citation validation, safe abstention, Docker, and CI; achieved **83.9%** Recall@5 and **92.9%** expected-source retrieval.

2. Financial Complaint Auto-Routing with NLP | Python • scikit-learn • TF-IDF • Linear SVM • PyTorch • DistilBERT

Jun 2026 - Aug 2026 [GitHub](#) | [Case Study](#)

- Built a leakage-safe eight-class CFPB complaint-routing study comparing a locked TF-IDF + Linear SVM benchmark with a frozen DistilBERT challenger; used group-aware 2024 splits with zero normalized-text overlap.
- Improved shared-2024 Macro F1 from 0.7671 to 0.7949 and evaluated both frozen models on a 30,156-row retrospective 2025 cohort using model-specific Auto-Route/Human Review policies.

3. NYC 311 Service Requests Lakehouse | Azure • Databricks • PySpark • SQL • Delta Lake

Mar 2026 - May 2026 [GitHub](#) | [Case Study](#)

- Built an Azure lakehouse pipeline using Azure Data Factory, ADLS Gen2, Databricks, Python/PySpark, SQL, and Delta Lake to process Bronze, Silver, and Gold layers.
- Implemented reusable ingestion, cleaning, SQL validation, and star-schema modeling to deliver analytics-ready fact tables, Gold marts, runbooks, architecture notes, and cloud execution proof.

4. Cloud Flight Fare Pipeline | AWS • Python • SQL • ECS/Fargate • S3 • Redshift • dbt

Feb 2026 - May 2026 [GitHub](#) | [Case Study](#)

- Built an end-to-end AWS batch pipeline using Python, Docker, ECS/Fargate, EventBridge Scheduler, S3, Redshift Serverless, SQL, and dbt to ingest and transform flight fare API data.
- Developed dbt staging models, dimensional marts, data quality tests, CI checks, architecture diagrams, runbooks, CloudWatch proof logs, and cost/secret safety documentation.

Experience

IT Intern - Fulton County | Jan 2024 - Apr 2024

- Standardized IT asset and network inventory records, improving data consistency and audit readiness.
- Reconciled records across multiple systems to support inventory reporting and internal documentation.

Floral Assistant - Kroger | Dec 2022 - Aug 2023

- Reconciled inventory, pricing, and shrink records to improve reporting accuracy and exception tracking.

Education

- **Master of Science (M.S.) in Information Technology, Data Analytics Concentration** - Middle Georgia State University, Expected Dec 2026 | GPA: 4.00
- **Associate of Applied Science (A.A.S.) in Website Design/Development** - Gwinnett Technical College, Dec 2024 | GPA: 3.96
- **Bachelor of Arts (B.A.) in Chinese Ethnic Language and Literature** - Guangxi University for Nationalities, China